FLOOD RISK MAPPING OF KOTA DISTRICT USING DIGITAL ELEVATION MODEL (DEM) AND GIS TECHNIQUES

A GIS-Based Hydrological and Terrain Analysis Project

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Project Category
Hydrological Analysis | Raster Analysis | Flood Risk Assessment

Software Used
QGIS 3.40 | GDAL Tools | GRASS GIS Tools | OpenStreetMap (OSM)

ABSTRACT

Floods are among the most frequent and destructive natural disasters affecting communities worldwide. They cause extensive damage to human life, property, infrastructure, agriculture, and the environment, leading to significant economic and social losses. With increasing urbanization, population growth, and climate variability, the frequency and intensity of flood events have increased in many regions. Therefore, identifying flood-prone areas and understanding the factors influencing flood occurrence have become essential for effective disaster management, risk reduction, and sustainable land-use planning. Advances in Geographic Information Systems (GIS) and remote sensing technologies have provided efficient tools for analyzing terrain characteristics and hydrological processes that contribute to flood susceptibility.

The present study focuses on flood risk mapping of Kota District, Rajasthan, using Digital Elevation Model (DEM)-based spatial analysis techniques in QGIS. The primary objective of the study is to identify potential flood-prone areas by analyzing topographic and hydrological characteristics of the district. Shuttle Radar Topography Mission (SRTM) DEM data were utilized as the primary dataset for terrain analysis. The DEM was processed to derive important parameters such as elevation, slope, flow direction, flow accumulation, and drainage network. These parameters play a crucial role in determining the movement and accumulation of surface runoff during rainfall events.

Elevation analysis was conducted to understand the variation in terrain height across the district, while slope analysis helped identify areas with low gradients that are more likely to retain water. Hydrological tools were applied to generate flow direction and flow accumulation rasters, which indicate the natural pathways and concentration of runoff. Based on a predefined flow accumulation threshold, a drainage network was extracted to represent the stream system of the district. Potential flood-prone zones were identified by combining areas of high flow accumulation with regions having slopes less than 5 degrees, as such locations are more susceptible to water stagnation and flooding.

The study resulted in the preparation of four thematic maps: Elevation Analysis Map, Slope Analysis Map, Drainage Network Analysis Map, and Flood Risk Analysis Map. The results reveal that low-lying areas situated near major drainage channels exhibit higher flood susceptibility compared to elevated regions. The generated flood risk map provides valuable spatial information that can assist government agencies, planners, and disaster management authorities in developing effective flood mitigation strategies and informed land-use decisions. The methodology adopted in this study is simple, cost-effective, and can be replicated for flood susceptibility assessment in other regions using freely available geospatial data and open-source GIS software.

Keywords: GIS, DEM, Flood Risk Mapping, QGIS, Slope Analysis, Flow Accumulation, Drainage Network, Kota District.

CHAPTER 1: INTRODUCTION

1.1 Background

Flooding is a recurring environmental hazard that affects millions of people every year. Rapid urbanization, changing climatic conditions, and improper land-use practices have increased the frequency and severity of flood events. Effective flood risk assessment requires understanding the interaction between topography, hydrology, and drainage systems.

Modern GIS techniques enable the integration of spatial datasets to model terrain characteristics and identify areas susceptible to flooding. Digital Elevation Models provide elevation information that can be used to derive hydrological parameters such as slope, drainage patterns, flow direction, and flow accumulation.

1.2 Need for the Study

Kota District experiences seasonal rainfall influenced by the southwest monsoon. The presence of the Chambal River and associated drainage networks makes understanding flood behavior essential. Identifying vulnerable zones can help authorities implement effective disaster management strategies and reduce potential losses.

1.3 Problem Statement

Flood-prone areas are often difficult to identify without detailed terrain analysis. Traditional methods require extensive field surveys, which are time-consuming and costly. DEM-based GIS analysis offers a rapid and cost-effective approach to flood susceptibility mapping.

1.4 Objectives

The objectives of this project are:

1. To prepare a Digital Elevation Model of Kota District.
2. To analyze elevation variation within the district.
3. To derive slope characteristics from DEM data.
4. To extract the drainage network.
5. To identify potential flood-prone areas.
6. To prepare thematic maps for flood risk assessment.
7. To demonstrate the application of GIS in disaster management.

1.5 Scope of the Study

The study focuses on terrain-based flood susceptibility assessment using DEM-derived parameters. The methodology can be applied to other regions with similar topographic conditions.

CHAPTER 2: STUDY AREA

2.1 Introduction

Kota District is located in the southeastern part of Rajasthan, India, and is one of the important administrative, industrial, and educational centers of the state. The district is situated within the Chambal River Basin and possesses diverse topographical and hydrological characteristics that make it suitable for flood risk assessment studies. The presence of the Chambal River and its tributaries significantly influences the drainage pattern and water movement within the region. Understanding the geographical and physical characteristics of Kota District is essential for analyzing flood susceptibility and developing effective disaster management strategies.

2.2 Geographical Location

Kota District lies between approximately 24°25' N to 25°51' N latitude and 75°37' E to 77°26' E longitude. It is bounded by Bundi District in the northwest, Baran District in the east, Jhalawar District in the south, and Chittorgarh District in the southwest. The district occupies a strategic location in the Hadoti region of Rajasthan and serves as an important transportation and commercial hub.

The district covers an area of approximately 5,446 square kilometers and consists of a combination of plains, valleys, plateaus, and riverine landscapes. These varied landforms contribute to differences in elevation, slope, and drainage characteristics across the district.

2.3 Climate

Kota District experiences a semi-arid climate characterized by hot summers, moderate winters, and a distinct monsoon season. Summer temperatures often exceed 45°C, while winter temperatures may fall below 10°C. The majority of annual rainfall occurs during the southwest monsoon period from June to September.

Rainfall plays a significant role in generating surface runoff and influencing flood conditions. During periods of intense rainfall, low-lying and poorly drained areas become vulnerable to water accumulation and flooding.

2.4 Topography and Drainage

The topography of Kota District is highly variable, with elevations ranging from approximately 177 meters to 511 meters above mean sea level. Higher elevations are primarily concentrated in the southwestern part of the district, while relatively lower elevations occur in the northern and central regions.

The district is drained mainly by the Chambal River, which serves as the principal river system. Several streams and tributaries form a well-developed drainage network that directs runoff toward the main river channels. The drainage pattern is predominantly dendritic, indicating natural development controlled by terrain characteristics.

2.5 Significance of the Study Area

Kota District was selected for this study because of its varied topography, extensive drainage network, and importance within the Chambal River Basin. The district experiences seasonal rainfall and possesses areas with low slopes and high runoff accumulation, making it suitable for flood susceptibility analysis. The results obtained from this study can support local authorities, planners, and disaster management agencies in identifying vulnerable zones and implementing effective flood mitigation measures.

CHAPTER 3: LITERATURE REVIEW

Flood risk assessment has become an important area of research due to the increasing frequency and impact of flood events worldwide. Researchers have extensively used Geographic Information Systems (GIS), Remote Sensing, and Digital Elevation Models (DEM) to analyze terrain characteristics and identify flood-prone areas.

Several studies have demonstrated the effectiveness of DEM-based hydrological analysis in flood susceptibility mapping. Smith et al. (2018) used GIS and DEM data to identify vulnerable flood zones and concluded that elevation and slope are critical factors influencing flood occurrence. Their study highlighted the importance of terrain analysis in disaster management planning.

Similarly, Kumar and Singh (2020) applied GIS techniques for hydrological modeling and drainage network extraction in river basins of India. They found that flow accumulation and drainage density play significant roles in understanding surface runoff patterns and flood behavior. Their research emphasized the usefulness of open-source GIS tools for watershed and flood analysis.

Patel et al. (2022) integrated DEM-derived parameters such as slope, elevation, and flow accumulation to prepare flood hazard maps. The study concluded that areas with low slopes and high runoff concentration are more susceptible to flooding.

The reviewed literature indicates that GIS-based terrain and hydrological analysis provides a reliable and cost-effective approach for flood risk assessment. These studies form the foundation of the present research, which applies DEM-based analysis techniques in QGIS to identify potential flood-prone areas in Kota District, Rajasthan.

CHAPTER 4: DATA AND SOFTWARE USED

4.1 Data Used:

4.1.1 Digital Elevation Model (DEM)

The primary dataset used in this study was the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM). The DEM provides elevation information for the Earth's surface and serves as the foundation for terrain and hydrological analysis.

4.1.2 Administrative Boundary Data

The administrative boundary of Kota District was used to define the study area and clip the DEM data. This ensured that all analyses were performed only within the district limits.

4.1.3 OpenStreetMap (OSM) Basemap

OpenStreetMap data were used as a reference layer for visualization and map preparation. The basemap provided contextual information such as roads, settlements, and geographical features that helped improve map readability and interpretation.

4.2 Software Used:

4.2.1 QGIS 3.40

QGIS 3.40, an open-source Geographic Information System software, was used for all spatial analysis and map preparation activities. QGIS provides powerful tools for raster processing, hydrological analysis, and cartographic visualization.

4.2.2 QGIS Processing Toolbox

The Processing Toolbox was used to perform various analytical operations such as:

- Slope generation
- Hillshade creation
- Flow direction analysis
- Flow accumulation analysis
- Stream extraction
- Flood risk mapping

The combination of high-quality spatial data and advanced GIS tools enabled accurate terrain analysis and effective identification of potential flood-prone areas within Kota District.

CHAPTER 5: METHODOLOGY

The methodology adopted in this study involves the use of GIS and DEM-based spatial analysis techniques to identify potential flood-prone areas in Kota District. The workflow consists of data preparation, terrain analysis, hydrological modeling, and flood risk mapping.

5.1 Data Preparation

The SRTM Digital Elevation Model (DEM) was downloaded and clipped using the administrative boundary of Kota District. The clipped DEM was then reprojected to an appropriate coordinate system for accurate terrain analysis.

5.2 Elevation Analysis

The DEM was used to generate an elevation map of the study area. Elevation values were classified into different categories to understand the topographical variation across the district.

5.3 Slope Analysis

A slope raster was derived from the DEM using QGIS terrain analysis tools. The slope map was classified into four categories: flat, gentle, moderate, and steep slopes. This helped identify areas with low gradients that are more susceptible to water accumulation.

5.4 Hydrological Analysis

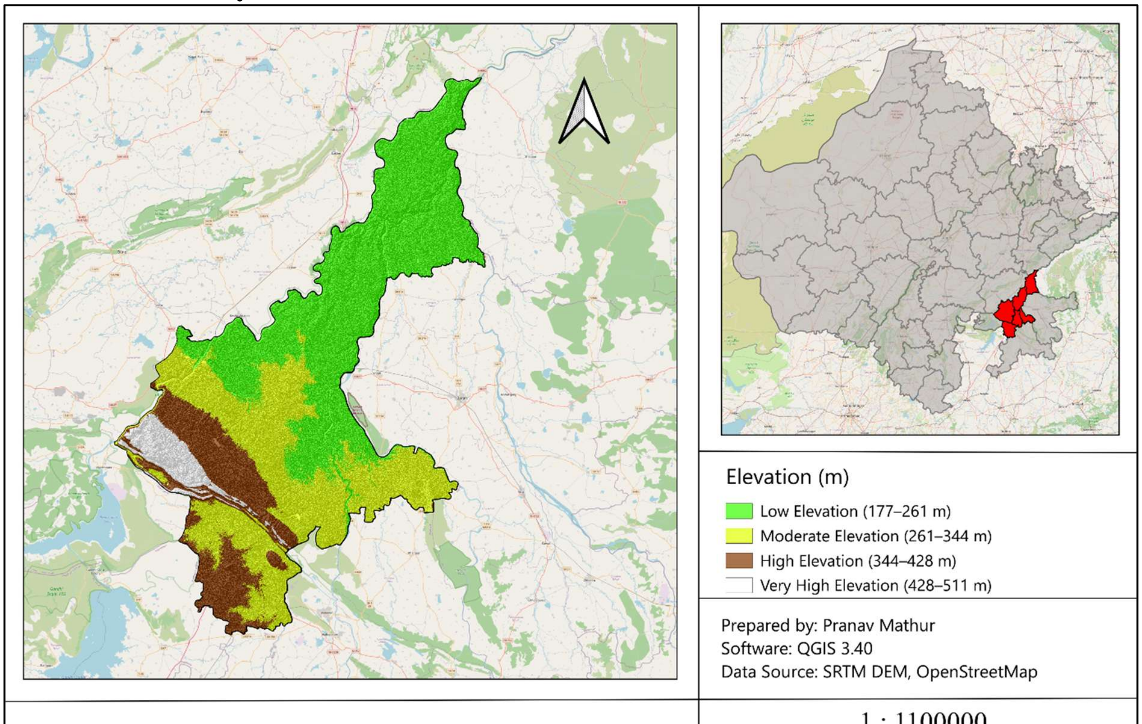
The DEM was processed to remove sinks and depressions. Flow direction and flow accumulation rasters were generated to model the movement and concentration of surface runoff. Based on flow accumulation values, the drainage network was extracted.

5.5 Flood Risk Mapping

Potential flood-prone areas were identified by combining hydrological parameters. Areas with Flow Accumulation > 1000 and Slope $< 5°$ were classified as flood-risk zones. Finally, thematic maps were prepared to visualize elevation, slope, drainage network, and flood susceptibility within Kota District.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Elevation Analysis



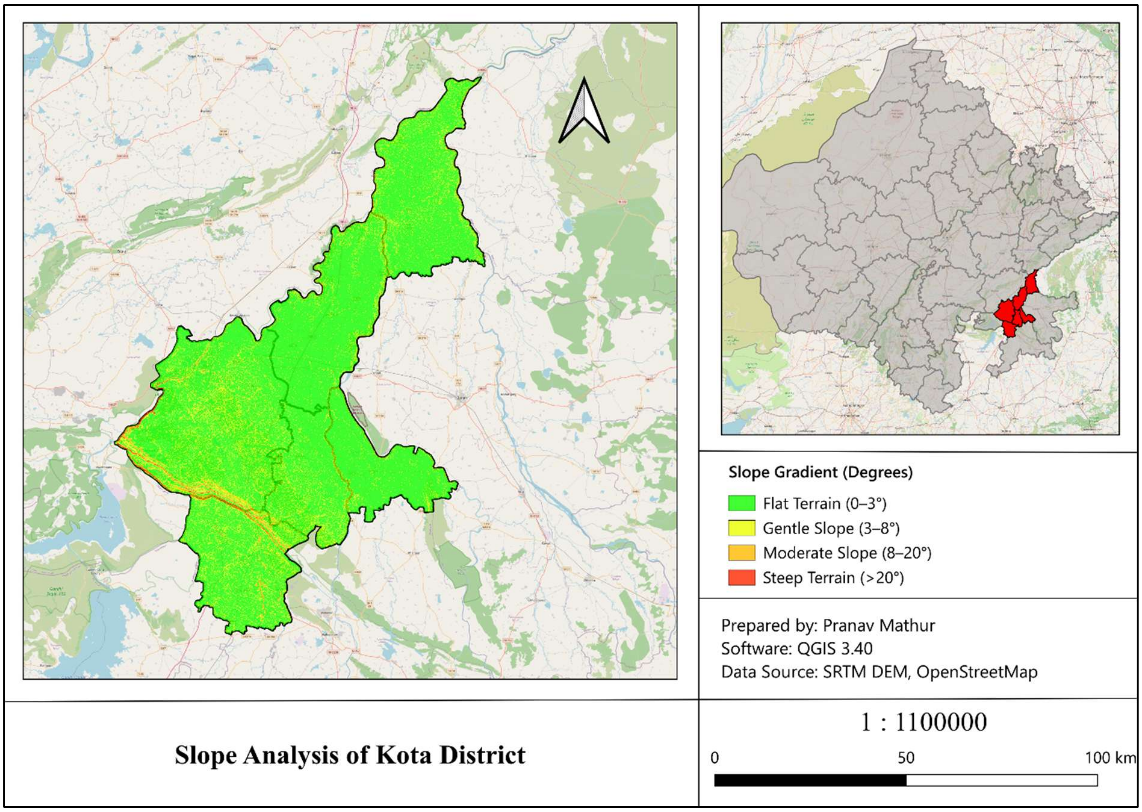
Elevation Analysis of Kota District

The elevation analysis was carried out using the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) to understand the topographical characteristics of Kota District. Elevation is one of the most important terrain parameters influencing surface runoff, drainage development, and flood susceptibility. The generated elevation map revealed that the elevation within the study area ranges from approximately 177 meters to 511 meters above mean sea level.

The analysis shows considerable variation in terrain across the district. Higher elevation zones are primarily concentrated in the southwestern part of Kota District, while relatively lower elevations are found in the northern, northeastern, and central regions. The elevation classes were categorized into low, moderate, high, and very high elevation zones to facilitate better interpretation of terrain characteristics. The southwestern region consists of elevated landforms, plateaus, and ridges, whereas the central and northern parts are comparatively flatter and lower in elevation.

Elevation plays a significant role in determining the movement of surface water. During rainfall events, runoff naturally flows from higher elevations toward lower-lying areas due to gravitational forces. As a result, elevated regions generally act as runoff-generating zones, while low-lying areas tend to receive and accumulate water from surrounding higher terrain.

6.2 Slope Analysis



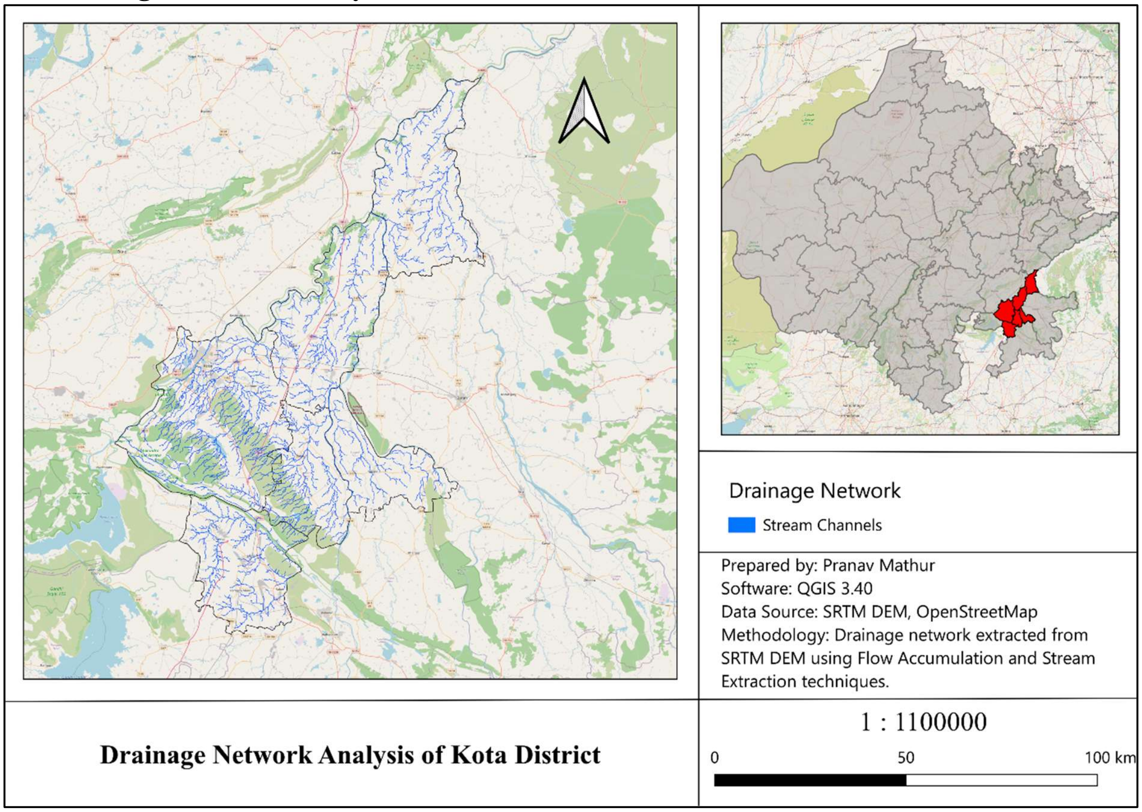
Slope analysis was performed using the Digital Elevation Model (DEM) to determine the degree of terrain steepness across Kota District. Slope is an important topographic parameter that influences surface runoff, soil erosion, drainage patterns, and flood susceptibility. The generated slope map classified the study area into four categories: flat terrain (0–3°), gentle slope (3–8°), moderate slope (8–20°), and steep slope (>20°).

The results indicate that a large portion of Kota District consists of flat and gently sloping terrain. These areas are predominantly distributed across the northern, central, and eastern parts of the district. Moderate and steep slopes are mainly concentrated in the southwestern region, where elevated terrain and escarpments are present. The spatial distribution of slope reflects the overall topographic variation within the district and influences the movement of water during rainfall events.

Slope plays a crucial role in determining the rate and direction of surface runoff. Low-slope areas facilitate slower water movement, allowing runoff to accumulate and remain on the surface for longer periods. Consequently, these regions are more susceptible to waterlogging and flooding, especially during heavy rainfall. In contrast, steeper slopes promote faster runoff and efficient drainage, reducing the likelihood of water accumulation and ponding.

When combined with elevation and hydrological parameters such as flow accumulation, slope helps identify areas that are more vulnerable to flooding. Therefore, the slope map serves as an essential component of the overall flood susceptibility analysis conducted in this study.

6.3 Drainage Network Analysis

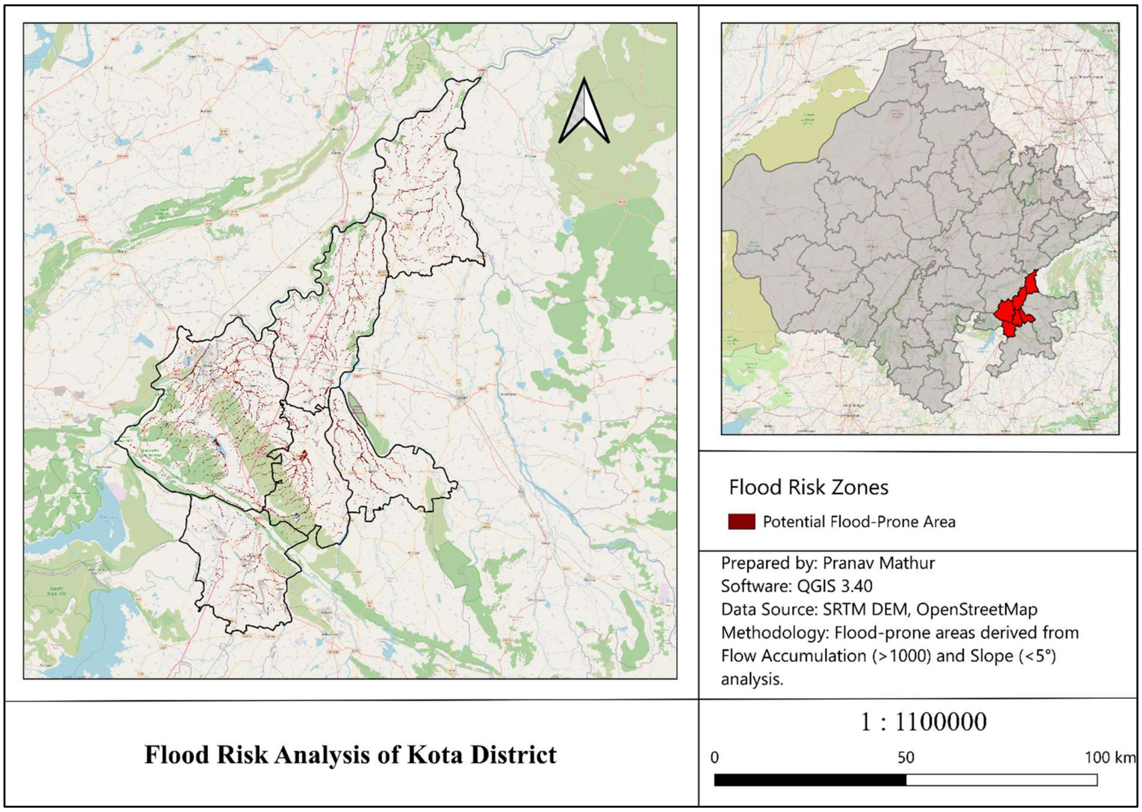


The drainage network of Kota District was extracted using flow accumulation analysis derived from the Digital Elevation Model (DEM). The resulting drainage map reveals a predominantly dendritic drainage pattern, which is characterized by a branching network of streams resembling the structure of tree branches. This type of drainage pattern typically develops in regions where the underlying geological structure is relatively uniform and does not significantly influence the direction of water flow.

The extracted drainage network represents the natural pathways through which surface runoff travels during rainfall events. The distribution of streams across the district provides valuable information about water movement, watershed characteristics, and runoff concentration. Areas located near major drainage channels are more likely to receive accumulated runoff from upstream regions, particularly during periods of intense precipitation.

The drainage network analysis is essential for understanding hydrological processes and identifying locations vulnerable to flooding. By indicating the direction and concentration of water flow, the drainage map supports flood risk assessment, watershed management, and planning activities aimed at reducing the impacts of flood-related hazards within Kota District.

6.4 Flood Risk Analysis



Flood risk analysis was carried out by integrating key hydrological parameters derived from the Digital Elevation Model (DEM). Potential flood-prone areas were identified using a combination of high flow accumulation values and low slope conditions, as these factors significantly influence the concentration and retention of surface runoff. Areas with a flow accumulation value greater than 1000 and a slope less than 5 degrees were classified as potential flood-risk zones.

The generated flood risk map highlights regions that are more likely to experience water accumulation during periods of heavy rainfall. The analysis indicates that potential flood-prone areas are primarily concentrated along major drainage channels and in low-gradient terrain. These locations receive runoff from larger upstream areas and often experience slower water movement due to gentle slopes, increasing the possibility of temporary flooding and waterlogging.

The results demonstrate the strong relationship between terrain characteristics and flood susceptibility. Identifying such vulnerable areas is important for disaster preparedness, land-use planning, and infrastructure development. The flood risk map can assist local authorities and planners in prioritizing monitoring efforts, implementing mitigation measures, and developing effective flood management strategies. Overall, the analysis provides valuable insights into the spatial distribution of flood susceptibility within Kota District and supports informed decision-making for sustainable development and risk reduction.

CHAPTER 7: CONCLUSION

The present study successfully demonstrated the application of Geographic Information Systems (GIS) and Digital Elevation Model (DEM)-based techniques for flood risk assessment in Kota District, Rajasthan. The project highlighted how freely available geospatial data and open-source GIS software can be effectively utilized to analyze terrain characteristics, hydrological processes, and flood susceptibility. By integrating elevation, slope, flow direction, flow accumulation, and drainage network analysis, a comprehensive understanding of the district's topography and hydrological behavior was achieved.

The analysis began with the preparation and processing of SRTM DEM data, which served as the foundation for all subsequent terrain and hydrological analyses. Elevation analysis revealed significant variation in topography across the district, with higher elevations concentrated in the southwestern region and lower elevations dominating the northern and central parts. Slope analysis further demonstrated that a large portion of the district consists of flat and gently sloping terrain, which influences runoff movement and water accumulation patterns.

Hydrological analysis provided valuable insights into the movement and concentration of surface runoff. Flow direction and flow accumulation tools helped identify natural drainage pathways and areas where water is likely to converge. The extracted drainage network revealed a dendritic drainage pattern that efficiently represents the natural flow system of the district. Understanding these drainage characteristics is essential for identifying areas that may experience increased runoff concentration during periods of heavy rainfall.

The flood risk assessment was carried out by combining flow accumulation and slope criteria. Areas with high flow accumulation values and low slopes were classified as potential flood-prone zones. The results clearly indicate that regions located near major drainage channels and low-gradient terrain are more susceptible to flooding and waterlogging. These areas tend to receive runoff from larger upstream catchments while also experiencing slower water movement, increasing the likelihood of temporary water accumulation during intense rainfall events.

The thematic maps generated in this study, namely the Elevation Analysis Map, Slope Analysis Map, Drainage Network Analysis Map, and Flood Risk Analysis Map, provide valuable spatial information for understanding the terrain and hydrological conditions of Kota District. These maps can support disaster management agencies, urban planners, engineers, and local authorities in identifying vulnerable areas, improving infrastructure planning, and implementing effective flood mitigation measures.

In conclusion, the study demonstrates that GIS and DEM-based spatial analysis techniques offer a reliable, cost-effective, and efficient approach for flood susceptibility assessment. The methodology adopted in this project can be replicated in other regions for preliminary flood risk evaluation and watershed analysis. Future studies may further improve the accuracy of flood risk assessment by incorporating additional datasets such as rainfall intensity, land use/land cover, soil characteristics, and historical flood records.

CHAPTER 8: FUTURE SCOPE

The present study successfully demonstrates the application of GIS and DEM-based techniques for preliminary flood risk assessment in Kota District. However, the accuracy and effectiveness of flood susceptibility mapping can be further improved by incorporating additional datasets, advanced analytical methods, and real-time monitoring systems. Future research can expand the scope of the study in several ways.

One important improvement would be the integration of rainfall data into the analysis. Rainfall intensity and duration significantly influence flood occurrence, and incorporating historical as well as real-time rainfall information would enhance the accuracy of flood risk prediction. Similarly, the inclusion of Land Use/Land Cover (LULC) data can provide insights into how urbanization, vegetation cover, agricultural land, and built-up areas affect runoff generation and water infiltration.

The study can also be improved by utilizing higher-resolution DEM datasets such as LiDAR or ALOS DEM. These datasets provide more detailed terrain information and can identify micro-topographic variations that are not captured by medium-resolution DEMs. As a result, flood-prone zones can be mapped with greater precision.

Another potential extension involves the use of advanced hydrological and hydraulic simulation models such as HEC-HMS and HEC-RAS. These tools can simulate rainfall-runoff processes, river flow dynamics, and flood inundation extents under different scenarios, providing more realistic flood hazard assessments.

Furthermore, the development of real-time flood monitoring and early warning systems using GIS, remote sensing, and IoT technologies can support timely decision-making during flood events. Such systems can help authorities monitor water levels, predict flood conditions, and issue warnings to vulnerable communities. These advancements would significantly enhance the effectiveness of flood management and disaster risk reduction strategies in the future.

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